SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I **Nevitha M/192521206** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Nevitha M

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

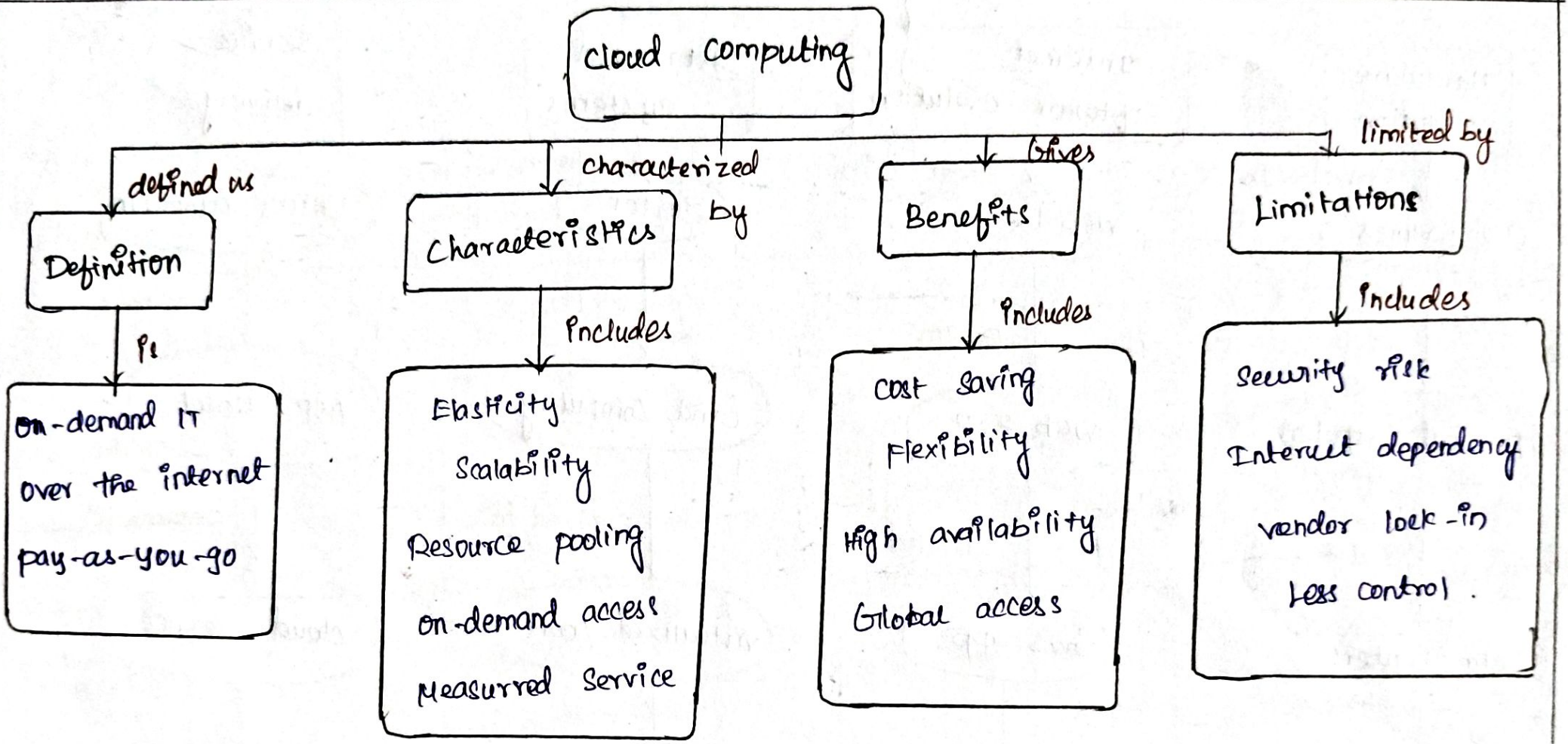
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

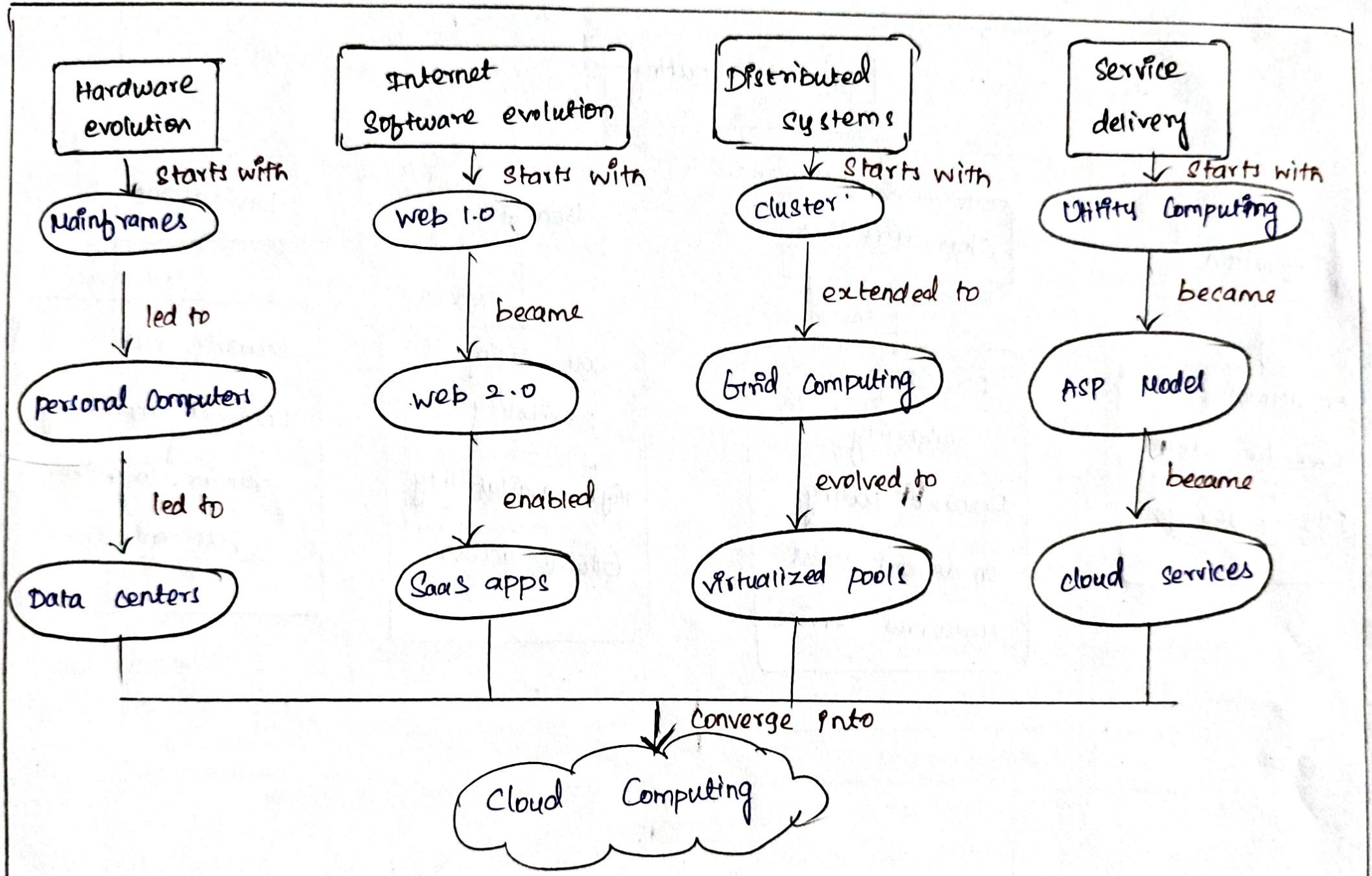
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

		identified			
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

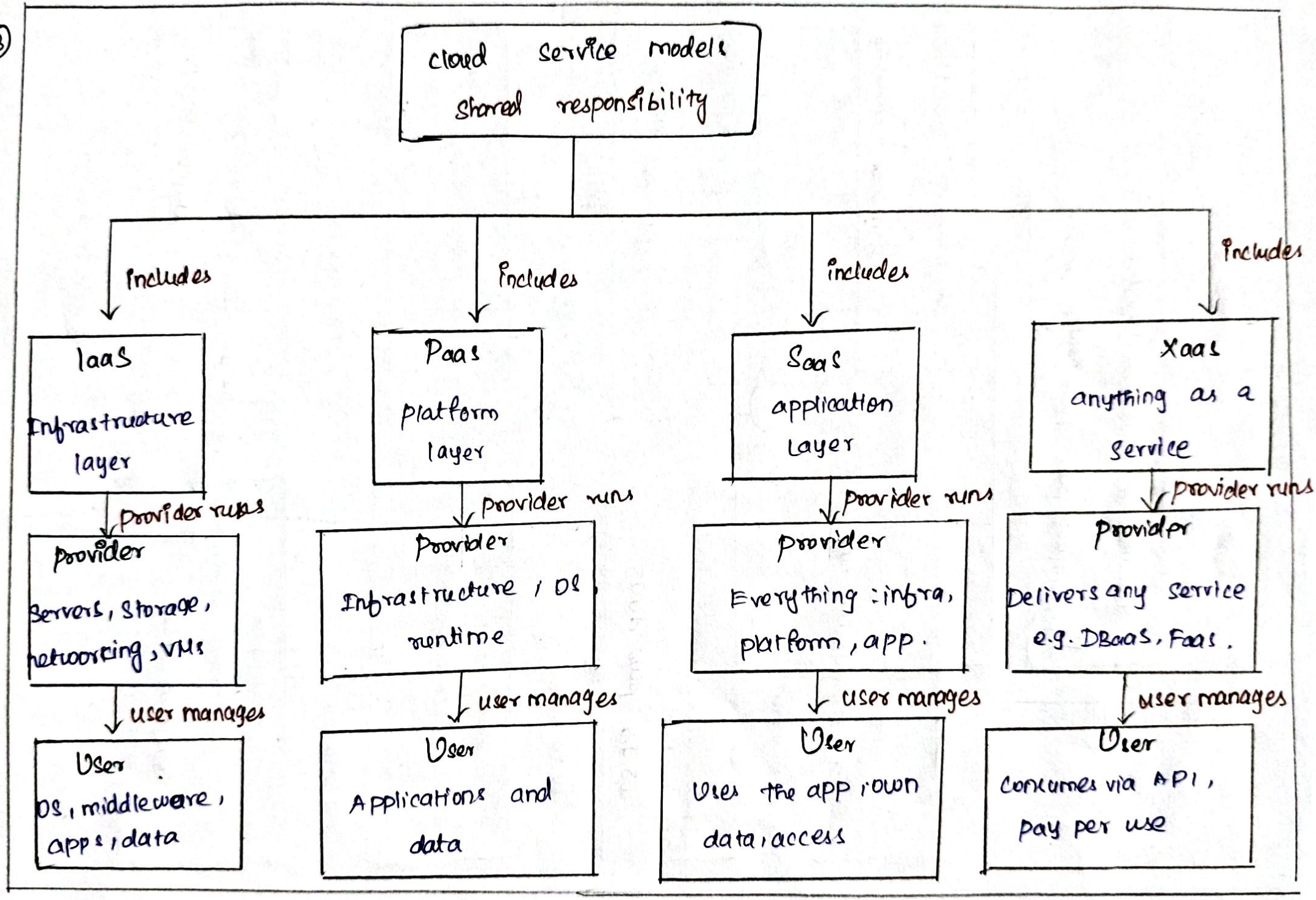
①


Cloud computing
defined as
characterized by
Gives
limited by
Definition
Characteristics
Benefits
Limitations
is
Includes
Includes
Includes
On-demand IT over the internet pay-as-you-go
Elasticity
Scalability
Resource pooling
on-demand access
Measurred Service
Cost saving
Flexibility
High availability
Global access
Security risk
Internet dependency
vendor lock-in
Less control.

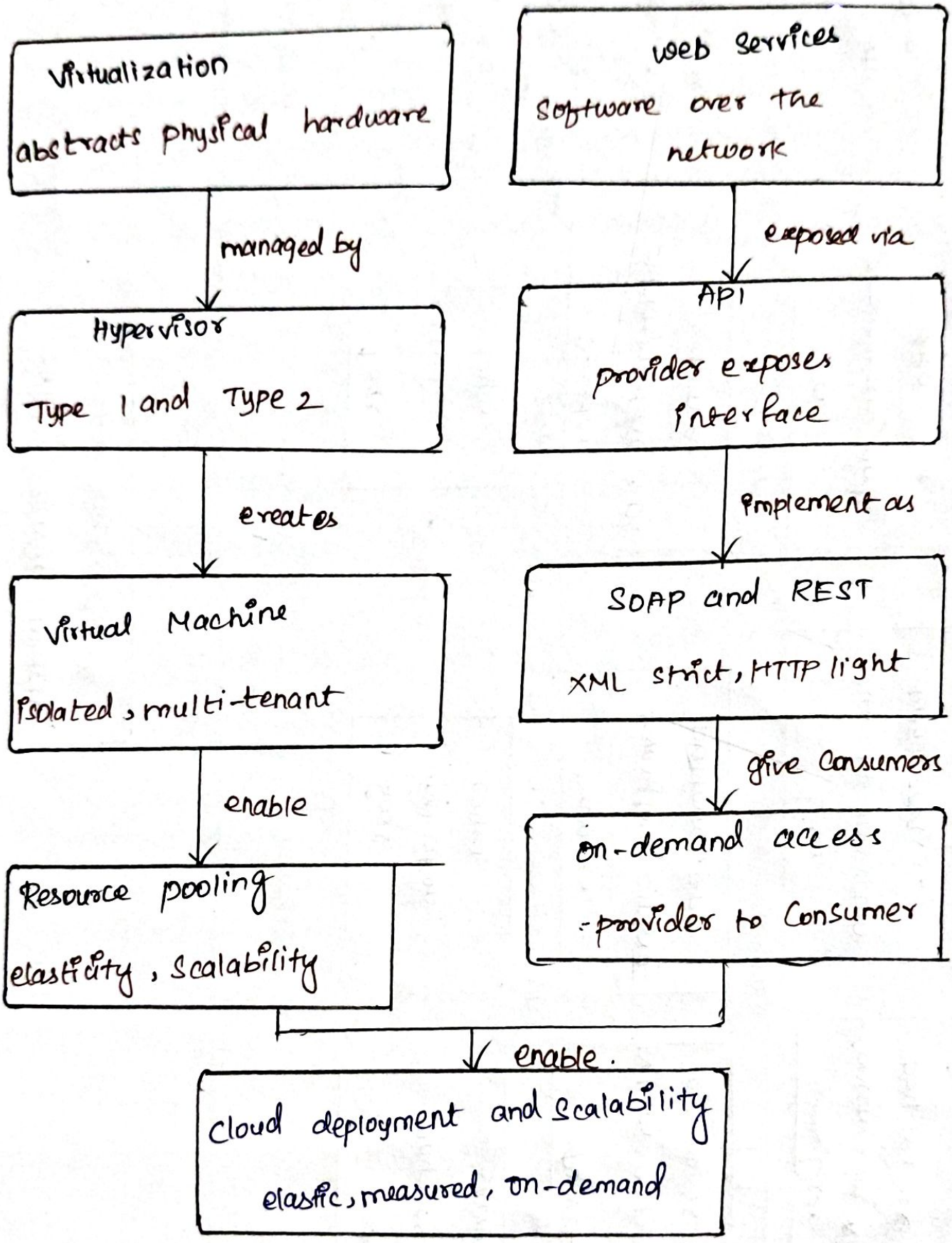
(2)


Hardware evolution
starts with
Mainframes
led to
Personal Computers
led to
Data centers
Internet Software evolution
starts with
Web 1.0
became
Web 2.0
enabled
Saas apps
Distributed systems
starts with
Cluster
extended to
Grid Computing
evolved to
Virtualized pools
Service delivery
starts with
Utility Computing
became
ASP Model
became
cloud services
Converge into
Cloud Computing

(3)


cloud service models
shared responsibility
Includes
Includes
includes
Includes
IaaS
Infrastructure layer
Paas
platform layer
Saas
application Layer
Xaas
anything as a service
provider runs
provider runs
provider runs
provider runs
provider
Servers, Storage, networking, VMs
provider
Infrastructure, OS, runtime
provider
Everything: infra, platform, app.
provider
Delivers any service e.g. DBaas, Faas.
user manages
user manages
user manages
user manages
User
OS, middleware, apps, data
User
Applications and data
User
Uses the app, own data, access
User
Consumes via API, pay per use

④


Virtualization
abstracts physical hardware
managed by
Hypervisor
Type 1 and Type 2
creates
Virtual Machine
Isolated, multi-tenant
enable
Resource pooling
elasticity, scalability
web services
Software over the network
exposed via
API
provider exposes interface
implement as
SOAP and REST
XML strict, HTTP light
give Consumers
on-demand access
- provider to Consumer
enable.
Cloud deployment and scalability
elastic, measured, on-demand